Consensus Is Not Evidence: It Decides Only Where No Rule Does

Abstract

Consensus adds no evidence in mathematics or science, because there a claim is settled by rules, a proof checked step by step or a test whose error rates are fixed in advance, and counting the people who accept the result adds nothing to it. Consensus decides only where no rule does. Whether an ambiguous mark reads as a 4 or a 9, whether a face is beautiful, whether a painting is art or a punishment is just: in such fuzzy categories the boundary is set by how competent judges draw it, and the agreement of many judges is the best answer there is. Those are the questions machine learning answers by training on many human labels, and the questions Socrates asked about justice, beauty and the good, which no definition has closed.

The two kinds of question call for different authority. Treating a rule-decided question as a vote, as "97 percent of scientists agree" does, moves science into the domain of judgment and asks a count to do the work of a test. This paper draws the line, shows where each kind of authority applies, answers the objection that non-experts have to rely on consensus, and proposes that scientific claims be reported by the evidence that would settle them, not by a head count.

Keywords: consensus; scientific evidence; expert testimony; vagueness; fuzzy categories; wisdom of crowds; social epistemology

1. Two kinds of question

Some questions are settled by rules and others by judgment. Is a number prime? Apply the test. Does a proof prove its theorem? Check each step against the rules

of inference (the valid patterns for drawing conclusions). Does a drug lower blood pressure more than a placebo? Run the trial, with the statistical test and its error rates (the chances of a false alarm or a miss) fixed before the data arrive. In each case the answer comes from a public procedure, and anyone who follows the procedure correctly gets the same answer.

Other questions have no such procedure. Is this handwritten mark a 4 or a 9? Is this face beautiful? Is this punishment just? The categories have clear centers and blurred edges, membership comes in degrees, and competent judges disagree at the margins (Zadeh, 1965). Nothing outside the pattern of judgments fixes where the edge lies.

Consensus means different things in the two cases. Where a rule decides, agreement is a report that many people applied the rule; the rule's result is the evidence, and the count adds nothing to it. Where no rule decides, the agreement of competent judges is what fixes the answer, and it is the best answer available.

Boaz Miller (2013) asks when a consensus is based on knowledge, Alvin Goldman (2001) how a novice can choose among experts, and Naomi Oreskes (2019) why the public should trust scientific consensus. This paper's contribution is to separate the domains. Where a rule decides, even a knowledge-based consensus is redundant with the evidence it rests on; where no rule decides, consensus is the answer itself. Its decisive premise is that in a rule-decided question the answer is fixed by applying the rule, whoever applies it. It rules out consensus as a reason within scientific argument. It leaves consensus a role as a pointer for outsiders and as the answer in fuzzy domains.

2. Where a rule decides, the count adds nothing

A theorem is true or false whatever anyone believes. Andrew Wiles announced a proof of Fermat's Last Theorem in 1993; a referee found a gap, and Wiles and

Richard Taylor closed it in 1994 (Wiles, 1995; Taylor and Wiles, 1995). Mathematicians accepted the result because the repaired proof checked, step by step. Had every mathematician believed the flawed version, the gap would still have been there.

Science works the same way with different rules. A measurement comes with its error bars; a hypothesis test comes with its error rates; a prediction can fail. The evidence is the data analyzed by procedures anyone can repeat. When a field's majority and the evidence part ways, the evidence wins. Most geologists rejected Alfred Wegener's continental drift (Wegener, 1915) until magnetic stripes on the seafloor showed the ocean floor spreading (Vine and Matthews, 1963). Physicians held that stress and acid caused stomach ulcers until Barry Marshall and Robin Warren showed the role of a bacterium (Marshall and Warren, 1984), work that won the 2005 Nobel Prize. Ignaz Semmelweis cut deaths from childbed fever (infection of new mothers after childbirth) by having doctors wash their hands, and the profession rejected him (Semmelweis, 1861). In each case a test settled the question and the count did not.

None of this makes experts useless. An expert is someone who can apply the rules, and an expert's report that the rules were applied carries weight. The value lies in the applied rules, which can be checked, and not in how many people report them. Consensus may tell an outsider where to look; it does not tell anyone what is true.

3. Where no rule decides, consensus is the answer

Fuzzy categories are where agreement does real work. The image collection that trained a generation of vision systems, ImageNet, settled each picture's label by the votes of many human labelers (Deng et al., 2009). There was no better method, because no rule says exactly where "dog" ends and "wolf" begins in a

blurry photograph; the label is what careful judges say. A model trained on millions of such labels learns the category as people draw it, which is why fuzzy questions are the ones machine learning handles best.

Pooling judgments also improves them. Francis Galton collected 787 guesses of an ox's weight at a country fair; their middle value, 1,207 pounds, came within one percent of the true 1,198 (Galton, 1907). The Marquis de Condorcet showed that if independent judges are each more likely right than wrong, a majority of them is more likely right than any one, and more so as the group grows (Condorcet, 1785). Where no one can apply the rule, many independent judgments are the best estimate available; once the ox is on the scale, the guesses add nothing.

The same holds for justice, beauty, art and the good. Socrates asked his companions for definitions that would work as rules: what piety is, what courage is, what beauty is. Each proposed definition failed against a counterexample, and the dialogues end without one (Plato, 1997: *Euthyphro*, *Laches*, *Hippias Major*). Where definition fails, the shared judgment of thoughtful people is what the words mean.

Fuzzy categories still follow rules, but rules too long for anyone to write down. Consensus is where such a rule is stored, spread across many judges, and a model trained on their judgments learns a compressed version of it. Where the rule is short enough to state and apply, as in arithmetic or a statistical test, the stored version is superfluous: apply the rule.

4. The cost of confusing the two

"97 percent of scientists agree" turns a physical question into a count. The best-known source of the figure counted the stances of paper abstracts: 97.1 percent of the 4,014 abstracts that took a position on human-caused warming endorsed it, and 66.4 percent of the 11,944 abstracts examined took none (Cook

et al., 2013). That is a measurement of what papers said. Whether and how much human activity warms the planet is settled, to whatever degree it is settled, by measurements and by models tested against them, with stated uncertainties. A count of stances adds nothing to those tests, and quoting the count in place of the tests invites the reply that counts can be manufactured.

The confusion has a general cost. An argument from consensus moves a question from the domain where rules decide into the domain where judgment decides, and judgment can be swayed by funding, fashion and politics in ways a test cannot. Science keeps its authority by pointing to the procedure anyone can repeat. The better report states the claim, the test that could have refuted it, and the result with its uncertainty. A reader who cannot run the test can still see that it exists and what it showed.

5. Objections

"Non-experts cannot check the evidence, so for them consensus is evidence." For an outsider, a consensus is testimony: evidence about what experts believe. Its weight comes entirely from the experts having applied the rules, independently, to the evidence. That makes it a pointer to evidence, not evidence of the same kind, and its strength depends on facts the count does not show: whether the experts judged independently, whether they could have been proved wrong, and whether dissent that holds up is rewarded. The more useful thing to give an outsider is the test and its result, which a count summarizes badly. A worked case shows where the weight lies. Two investigators, each right nine times in ten, who run the test independently and agree raise the odds from nine to one to eighty-one to one, about 98.8 percent. If the second copied the first, the agreement adds nothing, and the odds stay nine to one. The evidence was the second test, not the second name, and a head count cannot tell the two apart.

"Science has judgment calls too." It does: which research program looks promising, how to weigh conflicting studies before a decisive one exists, when evidence is enough to act on. Those are fuzzy questions, and there agreement describes a field's judgment. They should be reported as judgment, with the evidence still open, and not as the settled result of a test.

"Peer review and replication are consensus." Their value lies in checking work against rules: a reviewer finds the gap, a replication repeats the procedure. A paper is not more correct because more referees approved it; it is more trustworthy because more checks failed to find an error.

"Consensus usually tracks the evidence." When it does, the evidence does the work, and the count adds nothing to it. When it does not, the count misleads. Either way, the evidence is what should be reported.

6. Conclusion

Consensus is not evidence where a rule decides, because the rule's result is the evidence and the count adds nothing to it. Consensus decides only where no rule does: in fuzzy categories such as the label on a blurry image, the beauty of a face or the justice of a punishment, where the shared judgment of competent people is the answer. Mathematics and science stand on rules anyone can apply, and their claims should be reported by the tests that settle them. A head count belongs to the questions no rule can settle.

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